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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create a sample diabetes dataset
data = {
    'Pregnancies': [6, 1, 8, 1, 0, 5, 3, 10, 2, 4],
    'Glucose': [148, 85, 183, 89, 137, 116, 78, 115, 197, 110],
    'BloodPressure': [72, 66, 64, 66, 40, 74, 50, 0, 70, 92],
    'SkinThickness': [35, 29, 0, 23, 35, 0, 32, 0, 45, 0],
    'Insulin': [0, 0, 0, 94, 168, 0, 88, 0, 543, 0],
    'BMI': [33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 35.3, 30.5, 37.6],
    'DiabetesPedigreeFunction': [0.627, 0.351, 0.672, 0.167, 2.288, 0.201, 0.248, 0.
    'Age': [50, 31, 32, 21, 33, 30, 26, 29, 53, 30],
    'Outcome': [1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
}

diabetes_df = pd.DataFrame(data)

# 1. Display the first five rows
print("First Five Rows:")
print(diabetes_df.head())

# 2. Display the last five rows
print("\nLast Five Rows:")
print(diabetes_df.tail())

# 3. Display dataset information
print("\nDataset Information:")
diabetes_df.info()

# 4. Display the number of rows and columns
print("\nDataset Shape:")
print(diabetes_df.shape)

# 5. Display column names
print("\nColumn Names:")
print(diabetes_df.columns)

# 6. Check for missing values
print("\nMissing Values:")
print(diabetes_df.isnull().sum())

# 7. Display statistical summary
print("\nStatistical Summary:")
print(diabetes_df.describe())

# 8. Check duplicate rows
print("\nNumber of Duplicate Rows:")
print(diabetes_df.duplicated().sum())

# 9. Visualize the correlation between columns
plt.figure(figsize=(10, 6))
```

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sns.heatmap(diabetes_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Correlation Heatmap of Diabetes Dataset")
plt.show()

# 10. Visualize the outcome distribution
sns.countplot(data=diabetes_df, x='Outcome')
plt.title("Diabetes Outcome Distribution")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Count")
plt.show()
```

First Five Rows:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

Last Five Rows:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
5	5	116	74	0	0	25.6	
6	3	78	50	32	88	31.0	
7	10	115	0	0	0	35.3	
8	2	197	70	45	543	30.5	
9	4	110	92	0	0	37.6	

	DiabetesPedigreeFunction	Age	Outcome
5	0.201	30	0
6	0.248	26	1
7	0.134	29	0
8	0.158	53	1
9	0.191	30	0

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	10 non-null	int64
1	Glucose	10 non-null	int64
2	BloodPressure	10 non-null	int64
3	SkinThickness	10 non-null	int64
4	Insulin	10 non-null	int64
5	BMI	10 non-null	float64
6	DiabetesPedigreeFunction	10 non-null	float64
7	Age	10 non-null	int64
8	Outcome	10 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 852.0 bytes

Dataset Shape:

(10, 9)

Column Names:

Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
 dtype='str')

Missing Values:

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

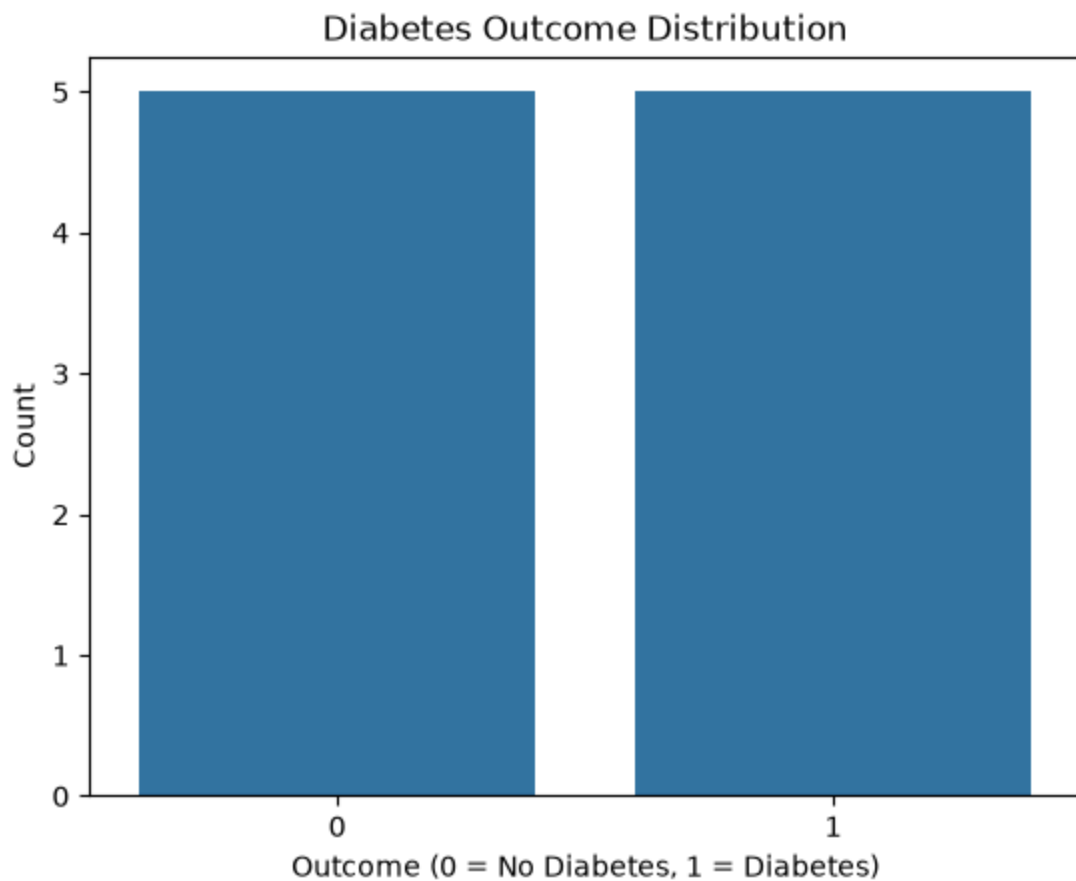
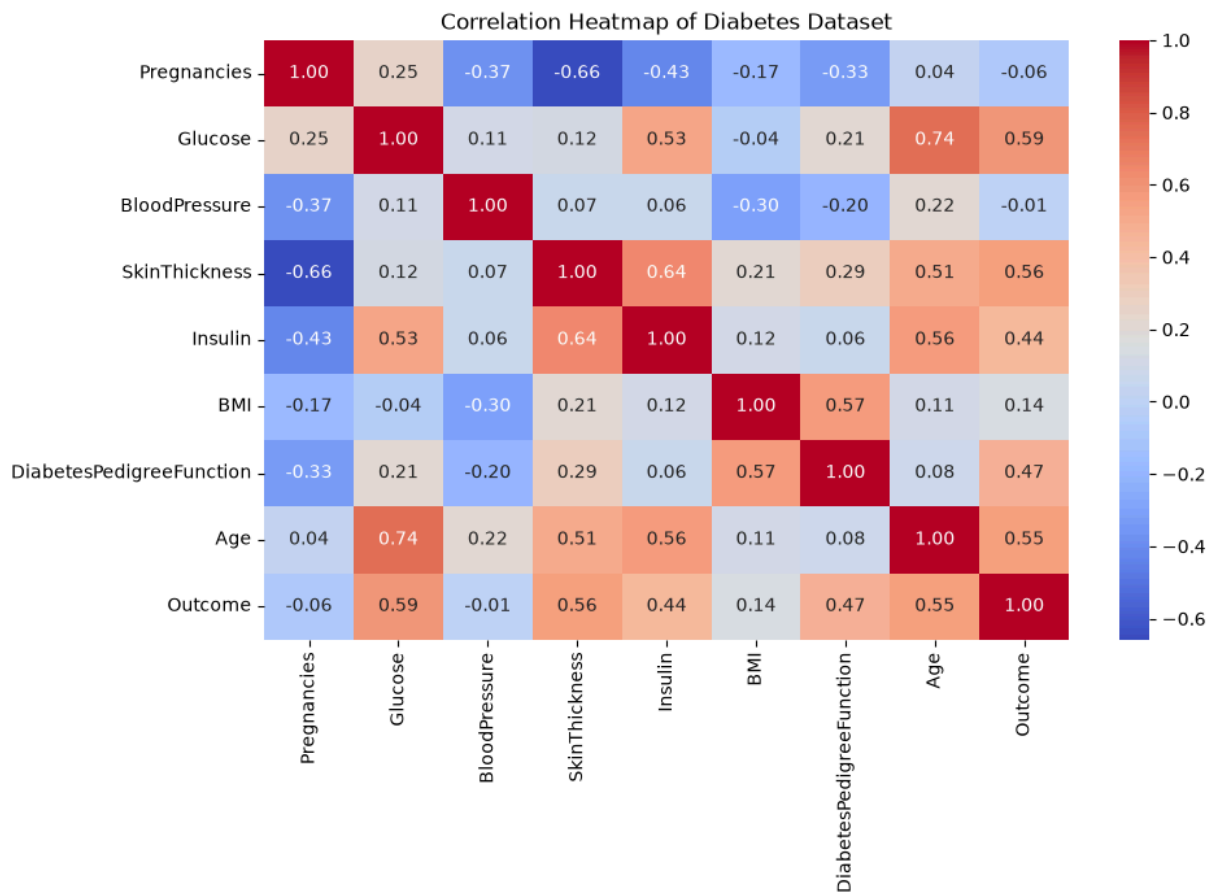
Statistical Summary:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	4.000000	125.800000	59.400000	19.900000	89.300000
std	3.265986	40.449694	25.051946	17.978073	169.937276
min	0.000000	78.000000	0.000000	0.000000	0.000000
25%	1.250000	94.250000	53.500000	0.000000	0.000000
50%	3.500000	115.500000	66.000000	26.000000	0.000000
75%	5.750000	145.250000	71.500000	34.250000	92.500000
max	10.000000	197.000000	92.000000	45.000000	543.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	10.000000	10.000000	10.000000	10.000000
mean	31.470000	0.50370	33.500000	0.500000
std	6.050537	0.65616	10.102255	0.527046
min	23.300000	0.13400	21.000000	0.000000
25%	26.975000	0.17300	29.250000	0.000000
50%	30.750000	0.22450	30.500000	0.500000
75%	34.875000	0.55800	32.750000	1.000000
max	43.100000	2.28800	53.000000	1.000000

Number of Duplicate Rows:

0



In []:

